(Please write your Enrollment Number) Enrollment No.________________

MID-TERM EXAMINATION

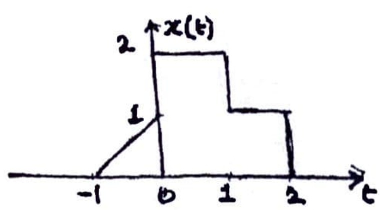
(FOR RE-APPEAR)

(Course Name: B. Tech. ECE) (Semester: 1st)

(March 2025) OFF LINE mode

Subject Code: BEC 106		Subject: Signal & Systems
Time: 1 ½Hours		Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) Define discrete time unit step & Unit impulse.		CO1
	b) Find whether the signal given by x (n) = 5cos (6n) is periodic or not?		CO1
	c) Write short notes on Dirichlet's conditions for Fourier Transform.		CO2
	d) Sketch the function x(t)= u(t)-2u(t-2).		CO2
	UNIT I		CO Mapping
Q2	Attempt any two parts	(5*2=10)	
	a) Are the systems represented by the following equations LTI system or not? y(n) = x(n-1) + 3x(n) + nx(n)		CO1
	b) Find the convolution of the following two signals and represent them graphically. $x(n) = \{\underset{\uparrow}{1}, 1, 0, 1, 1\}$ and $h(n) = \{1, \underset{\uparrow}{-2}, -3, 4\}$		CO1
	c) For given x(t). Draw x(-2t-2). 		CO1
	UNIT II		CO Mapping
Q3	Attempt any two parts	(5*2=10)	
	a) Find the Fourier transform of 1/(a+jt) if F.T $[e^{-at}u(t)] = 1/(a+jw)$		CO2
	b) Find the Fourier series representation of a periodic square wave f(t) = 1 for $0 < t < \pi$ = -1 for $\pi < t < 2\pi$		CO2
	c) The input and output of a causal LTI system are related by the differential equation, $d^2y(t)/dt^2+6dy(t)/dt+8y(t)=2x(t)$ Find the impulse response of the system?		CO2